Annealed Weak Router Supervision

Branch Specialization Phase 3 Toy Checkpoint

Branch specialization research project

2026-05-22

Checkpoint reason: the annealed-label experiment changes the next research step from broad regularizer sweeps to training-time trajectory measurement.

Current Question

Question. Does role-informative routing pressure only need to break symmetry early, or must it remain active through most of training for branch-level functional modularity to persist?

Setting. The conflict task makes the same query token require two incompatible role rules:

- local role: predecessor lookup, $y_i \to y_{i-1}$;
- induction role: successor lookup, $y_i \to y_{i+1}$.

Earlier checkpoints showed that unlabeled routing pressure, branch bottlenecks, and conflict alone did not produce role-aligned causal branch modularity, while weak role labels did.

Experimental Schedule

- Model: `weak_token_router`
- Task: `bidirectional_lookup`
- Branch head dimension: 64
- Training: 1600 steps, 5 seeds, equal local/induction loss weight
- Weak router-label loss: weight 0.05 while active

Label end step	50	100	200	400	800	1200	always
Fraction of training	.03	.06	.12	.25	.50	.75	1.00

The new flag is `--router-supervision-end-step N`; weak labels are active only for optimization steps $< N$.

Metric Glossary

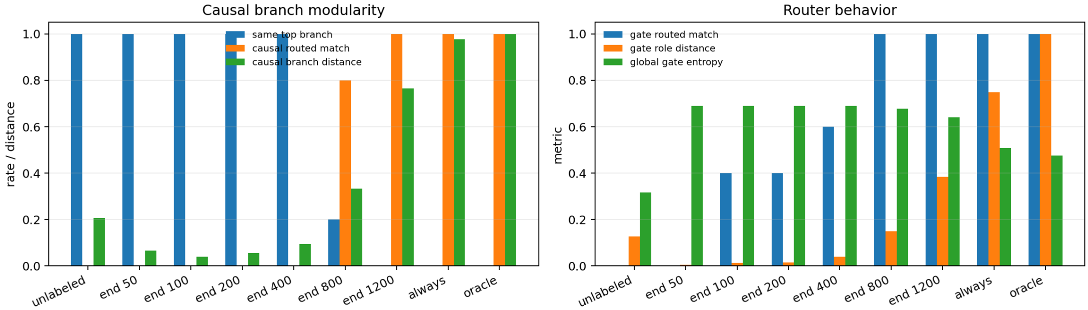
Same top branch Fraction of seeds where local and induction both have the same most-important branch under causal branch ablation.

Routed role match Fraction of seeds where local maps to branch 0 and induction maps to branch 1 under causal branch ablation.

Branch distance Average distance between the causal branch-use distributions for local and induction roles. Larger means stronger separation.

Gate routed match Whether router probabilities put the two roles on the intended branches. This is not enough by itself; the branch computation must also be causally separable.

Main Evidence



Left: causal branch modularity metrics. Right: router probability metrics. The discontinuity between end 400 and end 800 is the key empirical threshold in this toy setup.

Results Table

Condition	Label fraction	Same top	Routed match	Branch dist.
unlabeled entropy+balance	0.00	1.00	0.00	0.2069
end 50	0.03	1.00	0.00	0.0670
end 400	0.25	1.00	0.00	0.0945
end 800	0.50	0.20	0.80	0.3337
end 1200	0.75	0.00	1.00	0.7652
always label 0.05	1.00	0.00	1.00	0.9773
oracle route	–	0.00	1.00	1.0000

Readout. Brief early labels solve the task but do not preserve causal role separation. Labels through 75% of training preserve the top-branch split after removal, but with weaker separation than always-on labels.

Interpretation

Supported by this run

- Role-informative pressure does not need to remain active until the final step for a top-branch role split to persist.
- In this setup, it must last through a large fraction of training.
- Continuous role pressure gives stronger causal separation than late removal.

Not supported

- Brief role labels as a small early symmetry-breaking nudge.
- The stronger claim that structural branches plus task pressure alone reliably become functional branch modularity.

Project-Level Update

Best current toy conclusion.

Structural branch design is a substrate. In these toy runs, it did not by itself produce functional branch modularity, even with conflict, bottlenecks, entropy regularization, or load balancing.

The reliable ingredient observed so far is role-informative routing pressure. That pressure can be removed late with partial persistence, but short early exposure is not enough.

This reframes the next question:

What training signal makes structural modularity become functional modularity, and when must that signal be present?

Decision And Next Experiment

Decision. Stop broad sweeps of generic unlabeled router regularizers for this toy branch.

Next decisive experiment. Add training-time checkpointing and evaluate gate and causal branch separation at intermediate steps:

- before label removal;
- immediately after label removal;
- late in training after task loss has acted without labels.

Priority schedules: end 400, end 800, end 1200, always-on, and unlabeled entropy+balance.

Provenance

Source files

- scripts/toy_branch_isolation_intervention.py
- scripts/analyze_annealed_router_experiment.py
- doc/phase3_toy_annealed_router.md

Copied data in this deck folder

- data/annealed_summary.csv
- data/annealed_with_references.csv
- figures/annealed_router_comparison.png

Rebuild

- python3 -u scripts/analyze_annealed_router_experiment.py
- python3 .../compile_latex_deck.py presentations/2026-05-22-1709-annealed-router/annealed_router_checkpoint.tex